

YEAR: 2025
SEMESTER: 1
ASSESSMENT: Formative Assessment 2

SUBJECT NAME:	INTERNET PROGRAMMING
SUBJECT CODE:	INT316D
QUALIFICATION(S):	DPRS20/DPRSFO

PAPER DESCRIPTION: Computer Based **DURATION:** 3 HOURS **PAPER:** Only

SPECIAL REQUIREMENTS

- ☐ NONE
- ☐ NON-PROGRAMMABLE POCKET CALCULATOR
- ☐ SCIENTIFIC CALCULATOR
- ☐ COMPUTER ANSWER SHEET
- ☐ GRAPH PAPER
- ☐ DRAWING INSTRUMENTS

OTHER:

INSTRUCTIONS TO CANDIDATES: ANSWER ALL QUESTIONS

Send your work to the Lecturer's PC. Include the **project(s)**, **WAR** file, and the **Database (or tables)**.

TOTAL NUMBER OF PAGES INCLUDING COVER PAGE: 3

EXAMINER: MR V MEMANI

FULL MARKS: 80

MODERATOR:

TOTAL MARKS: 80

STUDENT TOTAL: ____

STUDENT %: ____

Question 1: Theory**[10]**

1.1 Discuss the following concept (refer to Note 1 below as a guide on the points that you need to include in your discussion):

1.1.1 JPA.

Note 1: In your discussion of JPA, address each of the following aspects:

- State what the acronym stands for. (1)
- Explain the purpose of JPA. (2)
- State what ORM stands for and its purpose. (2)
- Explain the concept of Persistence Provider. Also, give an example of a Persistence Provider. (3)
- State what the acronym JPQL stands for. (1)
- Explain the purpose of JPQL. (1)

Question 2: Practical - mini project**[70]****Problem statement**

Nyiko and Yinhla are entrepreneurial ICT students. Consequently, they have established a web development company called **TechnoByteSolutions**. The company specializes in developing **database-driven** web applications for clients.

TechnoByteSolutions has clinched a web development project from the **ICT Faculty**. The Faculty has established a working relationship with **industry** to benefit students. Part of the initiative is to expose third-year students to industry by participating in projects that solve real-world problems. The agreement is that each participating company will contribute three projects to the initiative. To manage the entire process, the Faculty needs a web application that will work with three roles (don't think security, think different links for different roles with different menu items), namely:

- Industry.
- Students.
- Faculty member.

Industry

The envisaged web application must allow the industry to **contribute** projects and **edit/update** their details whenever necessary. When contributing projects, the following information should be provided:

- Company name.
- Details of the three projects. For each project, a **title** and brief **description** must be provided.

The company information should be stored in a database together with a **timestamp**.

Students

The platform must also allow students to **view** the available projects contributed by the industry. In the next phase of the web application, students will be allowed to select their projects of choice, and the Faculty to do the final allocation. For now, viewing suffices.

Faculty member

A Faculty member must be able to view the following statistics:

- The total number of companies that have contributed projects.
- The total number of projects contributed.
- The total number of students who have viewed the projects.

Since this is a database-driven application, when creating the database, use the following information:

- Username: **app**
- Password: **123**
- Database name: **TechnoByteSolutionsDB**

To do

TechnoByteSolutions has been contracted by the **Faculty** to develop such a platform. You are a new developer at **TechnoByteSolutions**. As your first task, Nyiko and Yinhla allow you to prove your programming prowess by involving you in the development of the solution. Develop the required web application.

